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Phase-coded FMCW Laser Headlamp for Integrated Sensing, Communication, and Illumination

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***Abstract*—In this paper, we present an integrated sensing, communication, and illumination (ISCAI) system based on a phase-coded frequency modulated continuous wave (PC-FMCW) laser headlamp. The system unifies three key functionalities of gigabit-per-second data communication, high-resolution radar sensing, and adaptive driving beam (ADB) illumination. Phase-coded communication signals are embedded into frequency modulated continuous wave (FMCW to enable communication while maintaining sensing capability. To improve target perception, we develop a track-before-detect algorithm using a multidimensional Hough Transform (MHT). By decomposing the four-dimensional (4D) detection problem into two-dimensional (2D) projections, applying AND-logic fusion to suppress clutter, and using a rolling window for trajectory reconstruction, the algorithm achieves robust detection of weak and maneuvering targets. Simulation results demonstrate a 1 Gbps data rate, centimeter-level ranging accuracy (3.8 cm), and low tracking deviation (1.6787 units), highlighting the feasibility of deploying the proposed ISCAI system for intelligent connected vehicles.**

***Index Terms*—Integrated sensing, communication, and illumination, phase-coded FMCW, adaptive driving beam, multi-target tracking, multidimensional Hough Transform**

I. Introduction

Intelligent connected vehicles demand the co-design of sensing, communication, and illumination on a unified platform to ensure safety and efficiency in real traffic [1]. Sensing identifies road users and obstacles, while vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) connectivity enables cooperative, timely decision-making [2][3]. Adaptive headlights further enhance visibility by reshaping light distribution under varying scenes and weather conditions. However, implementing these functions with separate hardware inflates size, power, and cost, and introduces integration risks that undermine scalability and reliability [4]. To address this issue, integrated sensing and communication (ISAC) reuses spectrum and hardware within one stack and control plane, enabling sensing, communication, and illumination to share front-end components [5].

Recent research has explored optical approaches for integrating sensing and communication, achieving joint perception and data transfer through shared optical carriers. Phase-coded frequency modulated continuous wave (PC-FMCW) schemes have enabled lightweight joint signaling and high-resolution sensing in the optical domain [6][7]. To extend this concept to practical applications, researchers have further combined optical communication and illumination, enabling the same light source to support both data transmission and lighting functions. Among various implementations, light-emitting diodes (LEDs) and micro-LEDs are widely employed to realize visible light communication (VLC) while simultaneously serving as lighting sources [8][9][10]. However, LED-based systems suffer from limited modulation bandwidth and large beam divergence, whereas laser-based platforms provide much higher bandwidth and better directionality, enabling high-speed communication and precise sensing [11]. These advantages make laser-based architectures highly promising for integrated vehicular applications.

For optical ISAC systems, frequency modulated continuous wave (FMCW) processing provides range-Doppler information after coherent demodulation, forming the basis for trajectory detection of both stationary and moving targets. According to processing order, trajectory detection can be typically categorized into detect-before-track (DBT) and track-before-detect (TBD) approaches. DBT relies on thresholding individual frames, which leads to trade-offs between detection sensitivity and false alarms as well as range walk within a coherent interval [12][13]. In contrast, TBD accumulates weak target energy across multiple frames, enhancing detection of low signal-to-noise ratio (SNR) or maneuvering objects. Hough Transform (HT) can further improve robustness of TBD approaches to noise, data gaps, and non-uniform sampling [13][14][15]. However, its high-dimensional implementation is computationally expensive, requiring decomposition and projection strategies for real-time automotive deployment [16].

Motivated by these insights, we propose an integrated sensing, communication, and illumination (ISCAI) architecture that unifies optical hardware and coordinates resources at the physical layer [5]. The proposed system incorporates a laser-based headlamp to realize adaptive illumination while sharing the optical channel for sensing and communication. We further propose an embedded-oriented TBD algorithm to replace conventional single-frame DBT. A decoupled four-dimensional (4D) HT is introduced to perform multi-plane projection, per-plane voting, and rolling-window reconstruction to efficiently detect weak and maneuvering targets under low-SNR conditions. The proposed framework achieves ISCAI functionalities with real-time performance in automotive scenarios [7][9][13][16].

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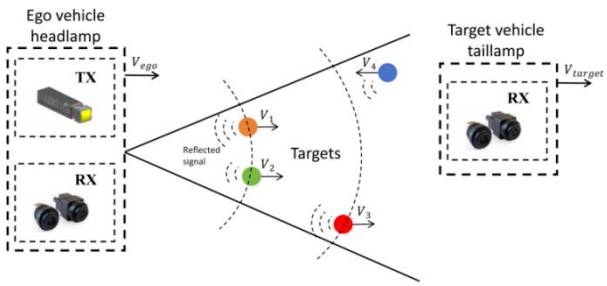

Fig. 1. Illustration of multi-target detection scenario.

II. PC-FMCW Laser Headlamp System Architecture

The proposed system embeds a PC-FMCW waveform onto a blue-laser carrier to jointly support sensing, communication, and illumination for intelligent connected vehicles (ICVs), as illustrated in Fig. 1. A coherently modulated laser applies phase coding to the FMCW chirp for data transmission. At the target vehicle, the receiver demodulates the data stream, while at the ego vehicle, the reflected beam is coherently detected and processed via digital signal processing (DSP) for ranging.

To ensure unified operation, the output optical signal of local oscillator (LO) adopts a linear FMCW structure and can be expressed as $s_{\mathrm{LO}}(t) = A_T \mathrm{e}^{j[2\pi f_c t + \pi\mu t^2]}$, where A_T, f_c, μ, and T_c denote the signal amplitude, carrier frequency, chirp slope, and chirp duration, respectively.

Data transmission is realized by embedding differential phase shift keying (DPSK) symbols within the FMCW phase domain, yielding the emitted optical signal $s_T(t) = A_T \mathrm{e}^{j[2\pi f_c t + \pi\mu t^2 + \varphi_d(t)]}$, where $\varphi_d(t)$ denotes the symbol-dependent phase. The phase term $\varphi_d(t)$ is piecewise constant over each symbol duration $T_s = 1/R_b$, producing differential phase shifts $\Delta\varphi \in \{0, \pi\}$ that encode the information bits. Phase transitions occur only at symbol boundaries and do not alter the chirp slope. The encoded signal propagates in free space where part of it is used for communication and the reflected portion enables ranging. As shown in Fig.1, at the targets, the signal is received for communication purposes, while a portion of the signal is reflected back and received at the transmitter of ego vehicle for ranging. In this context, the laser frequency $f_c = 193.4$ THz, chirp bandwidth $B = 10$ GHz, chirp period $T = 10$ μs, and data rate $R_b = 1$ Gbps are predefined parameters.

A. Sensing and Communication Subsystem

At the receiver of ego vehicle, the received echo, i.e., the reflected optical signal of a PC-FMCW signal, is converted into an electrical current by a photodetector and can be written as

$$s_{\mathrm{RX}}(t) = A_R e^{j\left(2\pi f_c (t-\tau) + \pi\mu (t-\tau)^2 + \varphi_d(t-\tau) + \varphi\right)}, \quad (1)$$

where $\tau = R/c$ represents the signal propagation delay and φ indicates a constant phase.

After mixing with the LO and low-pass filtering, the intermediate-frequency (IF) component is $s_{\mathrm{IF}}(t) = e^{j[-2\pi f_b t + \varphi_d(t-\tau) + \varphi]}$, where f_b is the beat frequency. To recover communication data, a fast Fourier Transform (FFT) extracts the carrier and differential demodulation yields DPSK symbols without relying on absolute phase, ensuring robust high-rate transmission alongside ranging.

For ego vehicle sensing with multiple targets, the received electrical signal is

$$s_{\mathrm{RX}}(t) = \sum_{p=1}^{P} A_p \, e^{j[2\pi f_c (t-\tau_p) + \pi\mu (t-\tau_p)^2 + \varphi_d(t-\tau_p) + 2\pi f_D^{(p)} t]}, \quad (2)$$

where τ_p and $f_D^{(p)}$ are the delay and Doppler of the p-th target.

After mixing and filtering, the low-frequency beat signals are retained. A group delay filter (GDF) [8] is applied in the frequency domain to cancel the phase coding, restoring the linear frequency modulation (LFM) structure and improving range-Doppler estimation accuracy. The GDF is characterized by the frequency-domain response $H_g(\omega) = e^{-j\omega\tau_g(\omega)}$, where the designed group-delay function $\tau_g(\omega)$ compensates for the phase perturbations introduced by DPSK coding. Subsequently, a two-dimensional (2D) FFT over fast-time and slow-time samples forms the range-Doppler map (RDM) as

$$X(m,k) = \sum_{n=0}^{N-1} x(m,n) \exp\left(-j\frac{2\pi}{N}nk\right), 0 \le k \le N-1, (3)$$

$$V(q,k) = \sum_{m=0}^{M-1} X(m,k) \exp\left(-j\frac{2\pi}{M}mq\right), 0 \le q \le M-1, (4)$$

where k is the index of range bins in the range domain and q is the Doppler frequency bin index, $X(m,k)$ denotes the range domain spectrum, $V(q,k)$ represents the range-Doppler spectrum after performing FFT across chirps. The power spectrum is $Z(q,k) = |V(q,k)|^2$. In the power domain, this work adopts the 2D cell-averaging constant false alarm rate (CA-CFAR) algorithm, which enables adaptive target detection while maintaining a constant false alarm probability.

Finally, adaptive detection is performed using CA-CFAR [17]. For each cell under test, local noise power is estimated from neighboring training cells by applying a threshold $T_{\mathrm{th}}(q,k) = \alpha \cdot \widehat{P}_{\mathrm{noise}}(q,k)$. If $Z(q,k) > T_{\mathrm{th}}(q,k)$, a target is declared, yielding robust detection with constant false alarm probability.

C. Illumination Subsystem

Adaptive driving beam (ADB) adjusts the headlamp output to maintain forward visibility while avoiding glare. A small portion of the modulated blue laser is used for sensing and communication, and the majority is converted to white light through a phosphor layer. The high-rate modulation is applied to the optical phase, and therefore does not introduce visible flicker. Guided by the camera and illumination control module, ADB reshapes the beam in real time. Following SAE J3069 [12], the illuminated area is divided into a non-glare zone and a transition zone to avoid discomfort for other road users.

Based on the defined zones, the system fuses camera data with PC-FMCW ranging to locate target vehicles and compute shading boundaries. The horizontal angular range is mapped

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into vehicle coordinates with lateral offsets and safety margins, and the shading interval is adjusted as $\Theta = [\theta_L - \frac{\Delta_y}{d} - \delta, \theta_R - \frac{\Delta_y}{d} + \delta]$, where d is the estimated range, Δ_y represents the lateral offset between the camera and the headlamp, and δ is a safety margin. A multidimensional light-intensity adjustment function $\mathcal{L}(\theta, d)$ is then constructed as

$$\mathcal{L}(\theta,d) = \begin{cases} 0, & \theta \in \Theta, d \le d_{\min}, \\ \lambda \dfrac{1-\cos\left(\pi \dfrac{d-d_{\min}}{d_{\max}-d_{\min}}\right)}{2}, & \theta \in \Theta, d_{\min} < d < d_{\max}, \\ 1, & \text{otherwise}, \end{cases} \quad (5)$$

where $[d_{\min}, d_{\max}]$ is the SAE-compliant distance threshold, and $\lambda \in [\lambda_{\text{low}}, \lambda_{\text{high}}]$ is the range of intensity adjustment. The attenuation function adopts a raised-cosine form for smooth transition.

III. Track Detection via Multidimensional HT

The proposed multi-target tracking algorithm is built upon HT, a robust parametric feature detection technique. In the HT framework, a line is represented in normal form as $\rho = x\cos\theta + y\sin\theta$, where ρ is the orthogonal distance from the origin to the line, and θ is the angle of the normal vector. Collinear points in the image space correspond to intersecting sinusoidal curves in parameter space, forming distinct peaks in a 2D accumulator matrix.

To detect linear tracks within three-dimensional (3D) spatio-temporal data (x, y, t) without incurring the computational burden of a full 3D HT, a multidimensional HT (MHT) is employed. Then 3D point cloud is projected onto three orthogonal 2D planes (xy, xt, yt), on each of which a standard 2D HT is performed independently. To improve sensitivity to weak tracks, a 3×3 mean filter is applied to each accumulator matrix before applying a voting threshold to extract significant peaks. For each detected line candidate (ρ, θ), original points are grouped if their perpendicular distance to the line satisfies $D = |x\cos\theta + y\sin\theta - \rho| < \tau$, where τ is a predefined distance threshold. A key innovation is the AND-logic fusion: candidate tracks are validated only when a sufficient number of common supporting points are present across all three projections. The main steps of the AND-logic fusion are summarized in Algorithm 1.

Algorithm 1: AND-Logic Fusion for Track Validation

Input: Point set $\{P_i\}$, minimum point threshold M
Output: Validated track segments $\{S_j\}$
For each projection plane $\pi \in \{xy, xt, yt\}$
  Obtain peak set C_π from the 2D Hough space
  Convert C_π to candidate line sets L_π by aggregating supporting points
For each triplet (s_{xy}, s_{xt}, s_{yt}) with $s_\pi \in L_\pi$
  $I \leftarrow$ intersection of supporting points in s_{xy}, s_{xt}, s_{yt}
  If $|I| \ge M$, add I to $\{S_j\}$ as a valid track segment
  return $\{S_j\}$

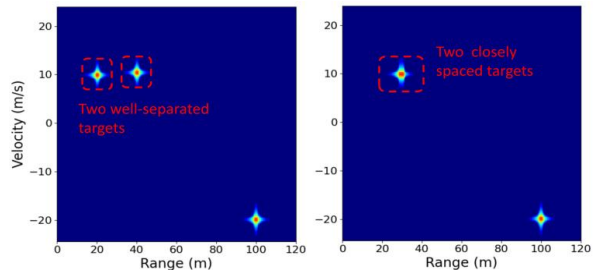

Fig. 2. Range-Doppler maps for multi-target scenarios: (a) well-separated targets yielding distinct peaks; (b) closely spaced targets producing overlapping peaks.

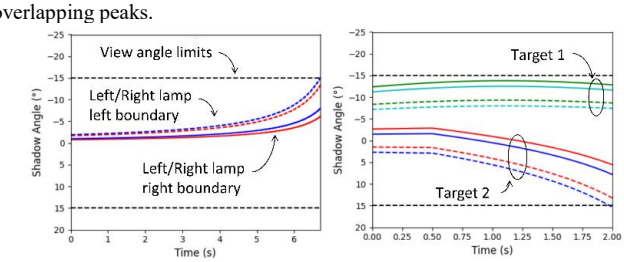

Fig. 3. ADB shadow angle adjustment under two traffic scenarios: (a) oncoming vehicle; (b) multiple preceding vehicles.

To further handle maneuvering or non-linear motion, the MHT operates within a rolling window of length W. Within each window, trajectories are approximated as piecewise linear segments. The detected segments T_{seg} are associated with global tracks T_k by minimizing a cost function

$$C(T_{\text{seg}}, T_k) = w_1 \cdot D_{\text{pos}} + w_2 \cdot D_{\text{kin}}, \quad (6)$$

which considers positional and kinematic consistency, where D_{pos} is the spatial distance between the segment centroid and predicted track position, D_{kin} is their velocity difference. If the minimum cost is below a stitching threshold, the segment is appended to the matched track; otherwise, a new track is created.

IV. System Validation and Performance Analysis

A. Sensing Performance

To quantify the sensing accuracy, we evaluate the Cramér-Rao Lower Bound (CRLB) for delay and Doppler estimation. For a single target, the observation vector is modeled as $\mathbf{y} = \mathbf{s}(\tau, \nu) + \mathbf{w}$, where $\mathbf{w} \sim \mathcal{CN}(0, \sigma^2 I)$ is a complex Gaussian noise vector with zero mean and variance of σ^2. Based on this model, the Fisher information matrix (FIM) for (τ, ν) is derived as

$$\text{var}(\hat{R}) \ge \left(\frac{cT}{2B}\right)^2 \frac{3}{8\pi^2 \gamma M T_c^2}, \text{var}(\hat{v}) \ge \left(\frac{\lambda}{2}\right)^2 \frac{3}{8\pi^2 \gamma {T_c}^2 M^3}. \quad (7)$$

where B is the sweep bandwidth, c is the speed of light, λ is the carrier wavelength, M is the number of chirps, T_c is the coherent integration time, and $\gamma = |A|^2/\sigma^2$ denotes the SNR.

For multiple targets, FIM includes off-diagonal terms indicating coupling of parameter estimates among these targets. For well-separated targets, these terms vanish and the performance approaches the single-target CRLB. For closely spaced targets, coupling increases estimation error and may even prevent resolution, as shown in Fig. 2; in contrast, well-

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separated objects yield distinct peaks, with a maximum ranging error of 3.8 cm observed for each target.

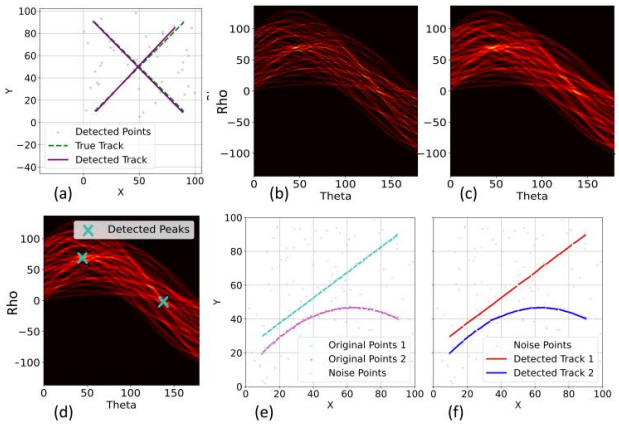

Fig. 4. Validation of the proposed MHT-based TBD algorithm: (a) raw data and detection results; (b) Hough space for the xy projection; (c) clustering result; (d) peak detection; (e) ground truth of linear and non-linear tracks; (f) final tracking output showing clutter suppression and trajectory reconstruction.

B. Adaptive Illumination Simulation

To mitigate glare from high-beam headlights, the proposed ADB subsystem creates dynamic shadow regions using multi-target information from the camera. Two representative scenarios are considered as follows.

(1) **Oncoming vehicle:** the host drives at 40 km/h, while an oncoming car approaches at 30 km/h with an initial separation of 150 m (Fig. 3a).

(2) **Multiple preceding vehicles:** the host drives at 50 km/h, following two vehicles in adjacent lanes with initial spacing of 30 m (Fig. 3b).

C. MHT Tracks Detection

To validate the proposed MHT-based TBD algorithm developed for the ISCAI system, we design two typical simulation scenarios. In the first scenario, a 100×100 2D space contains two linear tracks corrupted by Gaussian noise and random clutter. As shown in Fig. 4(a-d), the algorithm concentrates energy in the Hough space after smoothing, identifies distinct peaks, and groups points accordingly. The final AND-logic fusion effectively removes clutter with parameter errors of only 0.1251 and 0.2348 units.

In the second scenario, one linear and one non-linear tracks are embedded in dense clutter. Fig. 4(e-f) demonstrates that the algorithm accurately detected the linear track and reconstructed the curved trajectory by stitching local segments. Most clutter has been suppressed, and the mean deviation from ground truth is 1.6787 units, confirming the robustness of the rolling-window approach.

V. Conclusion

We proposed a unified PC-FMCW laser headlamp system that integrates sensing, communication, and illumination on a single platform. The design achieves gigabit-per-second data transmission and centimeter-level ranging accuracy, while enabling adaptive beamforming for safe illumination. To address weak target detection, a MHT-based tracking algorithm has been developed by combining decoupled voting, AND-logic fusion, and rolling-window association. The illumination results and the trajectory reconstruction results together verify the system's robust illumination control and accurate tracking performance. These results highlight the system feasibility for practical deployment in ICVs.

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